

REVIEW

Preparation and application of dielectric polymers with high permittivity and low energy loss: A mini review

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Abstract

This article reviews the preparation strategies of dielectric polymers with high permittivity and low energy loss, as well as their applications in the field of energy storage and intelligent sensors. We discuss the dependence of dielectric properties of polymer composites on different types of dielectric polarizations, including interfacial polarization, orientational polarization, ionic polarization, and electronic polarization, as well as the improvement of dielectric properties by synergetic effects of multiple polarizations. Due to the difficulties of achieving both high dielectric polarization (hence high permittivity) and low dielectric loss, we focus on the energy loss mechanism of different polarization types and issues in the preparation process. We also discuss current applications and future prospects of dielectric polymers in the field.

KEYWORDS

application, high permittivity, low energy loss, polymer, preparation

1 | INTRODUCTION

With the rapid growth of world economy and energy demand, the development of environment-friendly and cost-effective energy conversion/storage systems has become a hot research area of global attention.^{1–4} Besides energy storage devices,^{5,6} dielectric polymer materials have been widely used in the fields of artificial muscle,^{7–9} optoelectronics,^{10,11} pulsed power system,^{12,13} and sensor,^{14,15} which have aroused great interest of researchers in the field.

Dielectric constant (or permittivity) is the short name for relative dielectric constant (or relative permittivity),

which is the ratio of the permittivity of a substance to that of vacuum. Although permittivity and dielectric constant can be used interchangeably, the former is preferred in modern physics and chemistry. We will use permittivity throughout this review although dielectric constant is a common alternative in the literature. The permittivity of a polymeric dielectric varies within a wide range of values. Polymers with high permittivity have been widely used in energy storage and flexible sensing devices because of their excellent dielectric properties and flexibility.^{16–18} However, to achieve efficient energy conversion, an energy storage device is anticipated to exhibit a high energy density and a low energy loss. Applications of polymer dielectrics with high permittivity and low

Jie Qiu and Qun Gu contributed equally to this paper.

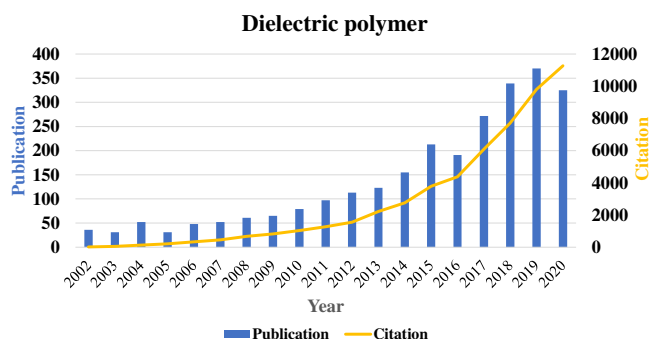


FIGURE 1 Number of publications and citations in dielectric polymer field from 2002 to 2020. The data were accessed from the search results in web of science by using keyword, that is, “dielectric polymer,” “high dielectric constant” and “low loss,” on February 8, 2022 [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.52367)]

dielectric loss in thin film capacitors in new energy vehicles or high-speed rail power equipment are reported and attracting interests.^{19–21} The data of publications and citations in dielectric polymer with high permittivity and low loss are shown in Figure 1, showing an increasing interest in this topic (the abnormal decreases in 2020 may be related to lower research activities during the COVID-19 pandemic).

The permittivity of a dielectric depends on its polarization, which is related to its microscopic structure and physical state. Dielectric polarization (or simply polarization) is caused by the shift of charges under an external field. Based on the mechanisms of how charges are shifted, polarizations can be roughly divided into three categories: distortion polarization, displacement polarization, and steering polarization.²² Distortion polarization is caused by the distortion of the electronic cloud of an atom or a molecule under an external field, producing a nonpermanent dipole moment, which leads to the so-called electronic polarization (for atomic species) or atomic polarization (for molecular species). In some cases, these two types are collectively referred to as Electronic Polarization (EP) (we use capital letters to distinguish it from the “conventional” electronic polarization [ep] occurring in an atomic species). Displacement polarization is caused by the displacement of charges. In a single-phase dielectric, the short-range displacement of bound charges leads to an instant ionic polarization; long-ranged displacement of free charges leads to a relaxation ionic polarization. Displacement of charges in a multi-component dielectric material is termed interfacial polarization, which often occurs at the filler-polymer interface in a polymeric composite. Steering polarization (also called orientational polarization or dipolar polarization) is caused by the realignment of permanent dipoles

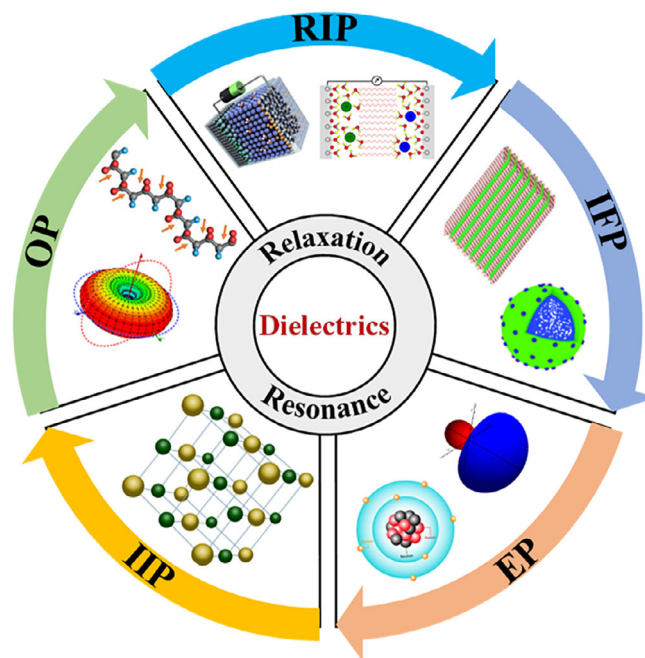


FIGURE 2 Microscopic description of dielectric polymers with different polarization types [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.52367)]

in the dielectric under the external field. Polarization can also be divided into two regimes based on the polarization rate (which determines the response time), namely, resonance regime (faster) and relaxation regime (slower). Electronic Polarization (EP) and the instant ionic polarization (IIP) belong to resonance regime, with EP faster than IIP. A polarization in the resonance regime occurs mainly in UV and IR frequency range and is temperature independent. Notably, EP contributes neither significantly to the permittivity of a dielectric, nor to the dielectric loss at all. We focus our discussion on other types of polarizations that affects both permittivity and dielectric loss. Interfacial polarization (IFP), relaxation ionic polarization (RIP), and orientational polarization (OP) belong to relaxation regime, with IFP the slowest and OP the fastest, in general. A polarization in the relaxation regime leads to a dielectric loss in radio frequency range; the energy dissipation occurs slowly and is temperature dependent. It is noteworthy to distinguish the two types of ionic polarization mentioned above. The IIP (in resonance regime) involves a charge displacement in the range of 0–0.05 nm in a dielectric polymer.²³ As we extend the discussion of ionic polarization to an energy storage device with dielectric-electrode connection regions, long-range charge displacement ($> \sim 10$ nm) is involved during RIP (in relaxation regime).²⁴ Despite different mechanisms, IFP and RIP share similar charge displacement and frequency range. Figure 2 illustrates

different types of polarizations occurring in polymer dielectrics.

The dielectric properties of polymers are generally characterized by permittivity, dielectric loss, and dielectric breakdown strength (DBS), all of which are attributed to the types of polarizations within the polymer.²⁵ Energy loss is the inherent loss of electromagnetic energy in the form of heat, which includes dielectric loss and discharge loss. It is inevitable that polarization, conduction, and discharge processes in a dielectric collectively can lead to a certain amount of energy loss. High polarization and low energy loss cannot be realized simultaneously. Therefore, one can only seek to achieve high permittivity at an acceptable cost of energy loss. In the meantime, high DBS, under a strong external alternating electric field, is desirable for a dielectric polymer used in energy storage devices, such as a capacitor.⁵ Although the vast majority of polymers with low dielectric loss and high DBS are widely used as dielectric and insulating materials in the electrical industry, their permittivity values are not sufficient for their applications in high energy equipment. Therefore, modified polymers²⁶ and polymer composites²⁷ are explored for optimized dielectric properties.

In this review, we discuss preparation and application of dielectric polymers with high permittivity and low dielectric loss, including nanocomposites, ion gels, and dipolar glass polymers. The merits and demerits of these polymer dielectrics will be discussed.

2 | RESONANCE-POLARIZATION-BASED POLYMER DIELECTRICS

Resonant dielectric polarization includes both EP (electronic and atomic polarization, 10^{14} – 10^{16} Hz) and IIP (10^9 – 10^{13} Hz). These polarizations could occur in all polymer systems. They are originated from the motion of electron clouds in an atomic, a molecular, or an ionic species induced by external fields, resulting in a deviation from their equilibrium positions. These polarizations occur at extremely high frequency ranges, with no dielectric losses in both the power frequency and RF ranges, which is a desirable feature for the enhancement of dielectric properties of materials. However, their contribution on permittivity is quite limited.

IIP is caused by the small relative displacement of cations and anions in the ionic crystal and generated by the molecule itself. It has a permanent dipole moment even without an external electric field. It is worth noting that the electric field is the main contributor to the permittivity. In polymer composites, it mainly exists in the doped inorganic crystals or ceramics.²⁸

Unlike ceramic materials, the atomic polarization in polymers is hardly enhanced further.²⁹ Since atoms are

connected by covalent bonds with well-defined bond lengths and bond angles, rather than ionic interactions, the contribution of atomic polarization in polymers to charge storage is minimal as compared to other types of polarization. According to electrostatic theory,³⁰ atomic polarization can be enhanced by replacing carbon atoms in polymers with atoms with a larger radius, including metallic and non-metallic atoms. In recent years, polymers containing zinc and tin moieties have been proposed for enhancing atomic polarization in polymers.³¹ The permittivity of polymers were improved compared to those of earlier discovered silicon atoms. In addition, enhanced atomic polarization can be achieved by increasing the polarity of polymers, as reported by related studies.³²

For electronic polarization, the permittivity of the polymer can be increased by further delocalization of electrons. Generally, the electronic polarization is 2–10 times that of the atomic. We know that although electrons are highly polarizable in semiconductor conjugated polymers,³³ electronic polarization is not suitable for high electric field applications because the electron conductivity in the matrix increases significantly at high fields. Also, the delocalization of the electrons usually decreases the band gap of the polymer. In general, although polarization in resonance regime can increase the permittivity of polymers to a certain extent, it is not sufficient for high-energy-density dielectric energy storage devices. Consequently, these polarizations are not the main contributors to optimizing the high storage electrical performance of polymer materials.

3 | RELAXATION-POLARIZATION-BASED POLYMER DIELECTRICS

Polarizations in relaxation regime take relatively longer time to occur, particularly IFP, which is followed by a discharge process lasting for hours or even years.^{23,34} The duration of discharge is closely related to the energy storage efficiency of the polymer composite, which may exhibit all three polarizations in this regime. While IFP contributes more to the permittivity comparing to RIP or OP, it frequently leads to serious energy loss.

3.1 | Dielectric polymers based on interfacial polarization

IFP occurs in response to an external field in the frequency range of 10^{-3} – 10^2 Hz. Due to the distinction between electrical properties of different components in a

heterogeneous system, space charges accumulate in large amount near (but not across) the interfaces between the polymer matrix and a certain component, thus triggering IFP. Typically, the IFP discharge works relative slow. Therefore, these dielectric polymers are not generally used in electrical energy storage devices requiring fast discharge, such as thin film capacitors. It is necessary to understand the relationship between IFP and dielectric loss, for example, the degree of IFP increases with increased number of accumulated space charges and the acceleration of charge migration, which leads to the increase of energy loss under the external electric field.

In recent years, polymer nanocomposites have attracted great interests as candidates of practical dielectrics for energy storage. Polymer nanocomposites are generally composed of a polymer matrix spiked with inorganic nanoparticles, such as ceramics with insulating properties,^{35,36} metals,^{37,38} or carbon particles with electrical conductivity.^{39–41} Although many new dielectrics have been reported recently, the development of IFP-dominated nanocomposites with high permittivity and low energy loss. A lot of research works focus on the dependence of the dielectric property of the composite on that of the nanofillers and their other relevant features, including shape, size, composition, and the dispersibility of nanofillers in the polymer matrix. The addition of certain nanofillers, such as BaTiO₃(BT) ceramic nanofillers,⁴² Zn nanoparticles,⁴³ and graphene oxide,⁴⁴ could increase the permittivity of the composite materials significantly. We discuss these materials in the next three sub-sections.

3.1.1 | Dielectric polymer composites with ceramic nanofillers

Ceramics are traditionally inorganic materials with high permittivity. They are brittle and processed at high temperatures. Therefore, ceramics cannot be used alone as dielectric materials in energy storage devices. To prepare a ceramic-polymer composite with high permittivity, a ceramic nanofiller is doped in a polymer matrix, which has a high volume fraction.^{45,46} Although the overall permittivity is dominantly attributed to the polymer, the ceramic filler can still effectively increase the permittivity of the composite.

The size of ceramic fillers affects the dielectric properties of composites significantly. Earlier research of polymer-based dielectric composites was mainly focused on ceramic fillers at micron scale. However, dielectric properties of these ceramic-polymer composites were compromised due to stress cracking and surface defects. Recent electronic industries call for nanoscale dielectric thin film materials to realize optimal energy storage with minimal volume fraction of the filler.^{47,48} Recently

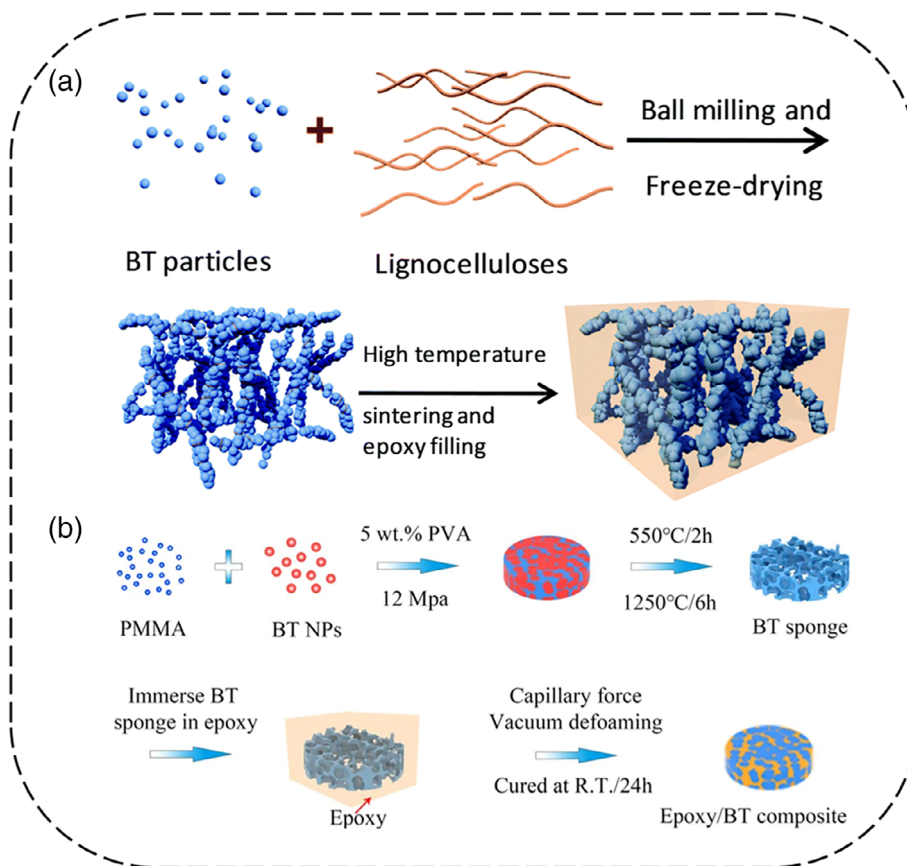
emerged thin film capacitors were made of polymer composites spiked with ceramic nanofillers. They showed desired dielectric properties, along with other merits, such as easy processability, low density, and excellent flexibility. Lu et al.⁴⁹ prepared self-supporting, flexible, and transparent ceramic-polymer nanocomposite films using Ba_{0.5}Sr_{0.5}TiO₃ (BST) nanoparticles as ceramic filler and P(VDF-CTFE) 91/9 mol.% (VC91) copolymer as matrix via a simple spin-coating process. The nanocomposite films exhibited high DBS, high permittivity, and low dielectric loss at high temperatures when the BST content was 40 and 50 wt%.

In addition to BST, nanoscale BT is a well-known ferroelectrics for the preparation of ceramic-polymer-based nanocomposite dielectrics with high permittivity values.^{50–52} BT has the perovskite type structure common for ferroelectrics, which contributes to its strong spontaneous polarization. Barium titanate/polyvinylidene fluoride (BT/PVDF) nanocomposites have excellent dielectric properties and are among the most widely studied composite systems. The dielectric responses of the nanocomposites strongly depend on the size of the BT fillers. Bi et al.⁵³ prepared a BT/PVDF nanocomposites with 92.3, 17.8, and 5.9 nm BT particles. Increased filling amount has improved both the breakdown strength and the dielectric properties of the nanocomposites. In another research, polymer composites based on a three-dimensional interconnected BT network were prepared by injecting epoxy solution into the porous structure,⁵⁴ as shown in Figure 3a. It was found that this synthetic strategy greatly enhanced the contribution of BT to the energy storage properties of the composites. The construction of the 3D-BT scaffold is proved to be effective in enhancing the IFP of the composites, which has greatly increased the permittivity and energy density.

The past 3 years have witnessed significant new breakthroughs in the study of dielectric properties of BT polymer-based ceramic nanocomposites. BT sponge-filled polymer composites with high permittivity and adjustable porosity using poly(methyl methacrylate) (PMMA) microsphere arrays were produced by Jian et al.⁵⁵ The preparation process is shown in Figure 3b. The BT/epoxy resin sponge composite seems to be a promising material for electrostatic energy storage capacitors. Although BT has a high permittivity, it needs to be sintered at high temperature before preparation of ceramic-polymer composites, which leads to poor mechanical property and low breakdown voltage of the composites formed.

To achieve a sustainable development, environment-friendly ceramic-polymer composites based on SrTiO₃(ST), BT, Bi_{0.5}Na_{0.5}TiO₃(BNT), AgNbO₃(AN), K_{0.5}Na_{0.5}NbO₃(KNN), and BiFeO₃(BF) have been studied.^{56–61} Bismuth ferrite (BiFeO₃, BF)-based lead-free ceramics used in pulsed power capacitors have attracted considerable attention because of their advantages such as large spontaneous polarization

FIGURE 3 Schematic diagram of composite preparation. (a) Schematic diagram of 3D BT/epoxy composite preparation.⁵⁴ Copyright 2017, the Royal Society of Chemistry. (b) Preparation of spongy BT 3D structure by using PMMA sphere array as a template. The BT/epoxy sponge composites were prepared by capillary force and vacuum defoaming method.⁵⁵ Copyright 2020, springer nature [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]



(hence high permittivity), low sintering temperature, and nontoxicity. However, low energy density due to higher dielectric loss and instability of these ceramics do not favor their feasible applications. Yan et al.⁶² successfully designed a new bismuth ferrite based lead-free ceramic, (1-x)BFBT-xNBN. They added NBN to decrease remnant polarization with improved maximum polarization and DBS by optimizing the polarization under an electric field with a suitable field strength. This work demonstrates that (1-x)BFBT-xNBN not only exhibits favorable energy storage properties but also shows good mechanical stability, which is expected to be an ideal material for pulsed power capacitors. Ceramic-polymer composites prepared with traditional ceramic fillers do have high permittivity, but their poor mechanical properties and obvious dielectric loss limited further applications in energy storage devices. Therefore, it is imperative to develop new ceramic fillers to improve the properties of ceramic-polymer composites, for which researchers have shown extensive interest.⁶³

3.2 | Metal-polymer dielectric nanocomposites

Metallic nanofillers in appropriate amount can improve dielectric property of polymer composites.²⁴ When

conductive metal nanofillers are added to a polymer matrix, space charges (electrons and holes) would accumulate at the metal-polymer interfaces to trigger IFP occur. Although the permittivity of metal-polymer composites is generally high, many unsolved issues are still existed. Conduction loss is caused by movement of charges that experiences resistance or frictions, which leads to energy dissipation in the form of heat. In theory, as the amount of conductive fillers in a polymer increases, the distance between the filler particles became smaller. As the doping amount approaches a threshold (called percolation threshold), the conductive filler particles are so close to one another as to form a conductive network structure (also known as percolation network), because of the electron tunneling effect. Once the amount of filler surpasses the percolation threshold, the composite will be transformed into a conductor. Percolation must be avoided in the preparation of dielectric composites.

The dielectric properties of metal-filled polymer are commonly unstable and irreproducible because of non-uniformity of particle sizes and inter-particle distances. Proper amount, uniform size, and homogeneous dispersion of metal fillers are critical to avoid unwanted percolation and enhance the performance of the dielectric materials. To achieve these, one can modify the surface structure of metal fillers to improve filler-polymer

interfacing. In addition to dielectric loss during polarization, polymer composites based on conductive nanofillers suffer from regional inter-particle conduction loss where particles are too close to one another, an additional source of energy loss not seen in ceramic fillers. Homogeneous dispersion of metal fillers in the matrix is highly desired to reduce energy loss.

A lot of research has been done to address this issue. One investigation is about the effect of core-shell structure of the metal-semiconductor, Zn-ZnO, on the dielectric properties of polyvinylidene fluoride (PVDF) composites.⁶⁴ The increased permittivity of the composite arises from the dual-interfacial polarization at the metal-semiconductor and the semiconductor-insulator interfaces. In addition, the presence of a ZnO shell between the zinc core and the insulator (PVDF matrix) promotes the dispersion of filler particles in the matrix, resulting in low dielectric loss. At the same time, the dielectric properties of the composite have been further optimized by adjusting the relative thickness of the semiconductor shell layer.

Zhang et al.⁶⁵ proposed an approach to combine the processes of solution casting and hot pressing was used to prepare homogeneous nickel-polymer nanocomposites. Two systems, P(VDF-TrFE)/70/30 mol.% and P(VDF-CTFE)/88/12 mol.%, were used to study the dielectric properties of the composites, filled with nickel nanoparticles with different volume fraction ranging from 0 to 60%. It is found that the two nanocomposite systems have very similarly high percolation thresholds and high permittivity (~ 1000 at 100 Hz). Unfortunately, both nanocomposite systems have exhibited high dielectric losses in low-frequency electric fields. Mixed nanoparticles of ionic compound and metal have been used to improve dielectric properties. As reported by Morsi et al.,⁶⁶ LiBr and biosynthesized silver nanoparticles (AgNPs) with an average particle size of 2–30 nm were casted and incorporated into the matrix of polyethylene oxide/polyacrylamide (PEO/PAM) blends. The composites exhibited enhanced permittivity with reduced dielectric loss.

Recently, metal-polymer dielectric composites with good mechanical and dielectric properties are becoming popular in the area of energy storage devices. Publications investigating the properties of particulate metal-polymer composites have been growing. Dielectric materials based on poly(*n*-butyl methacrylate)-coated, liquid metal core-shell structured nanodroplet (EGaIn@PBMA, “EGaIn” stands for “Eutectic Gallium Indium alloy”) were fabricated using in situ free radical polymerization method,⁶⁷ as shown in Figure 4. The flexible nanoparticles can be directly hot-pressed into the matrix to produce the nanocomposites, which exhibit ultra-high rate of elongation ($>400\%$), flexibility, and excellent dielectric properties.

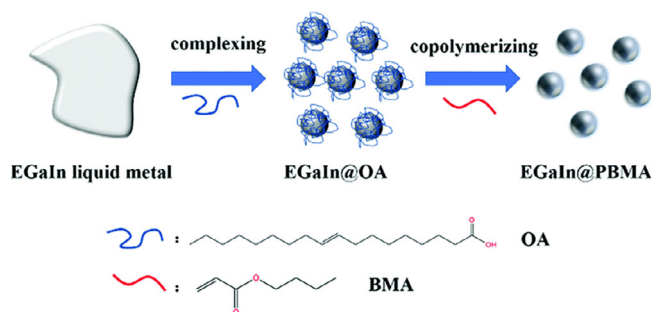


FIGURE 4 Mechanism of synthesis of nanodroplets of core-shell structure EGaIn@PBMA.⁶⁷ Copyright 2020, Royal Society of chemistry [Color figure can be viewed at wileyonlinelibrary.com]

Flexible metal-polymer composites have broad application prospects in the field of flexible equipment, such as wearable/flexible sensing and energy conversion devices.

3.2.1 | Carbon nanomaterial-polymer dielectric composites

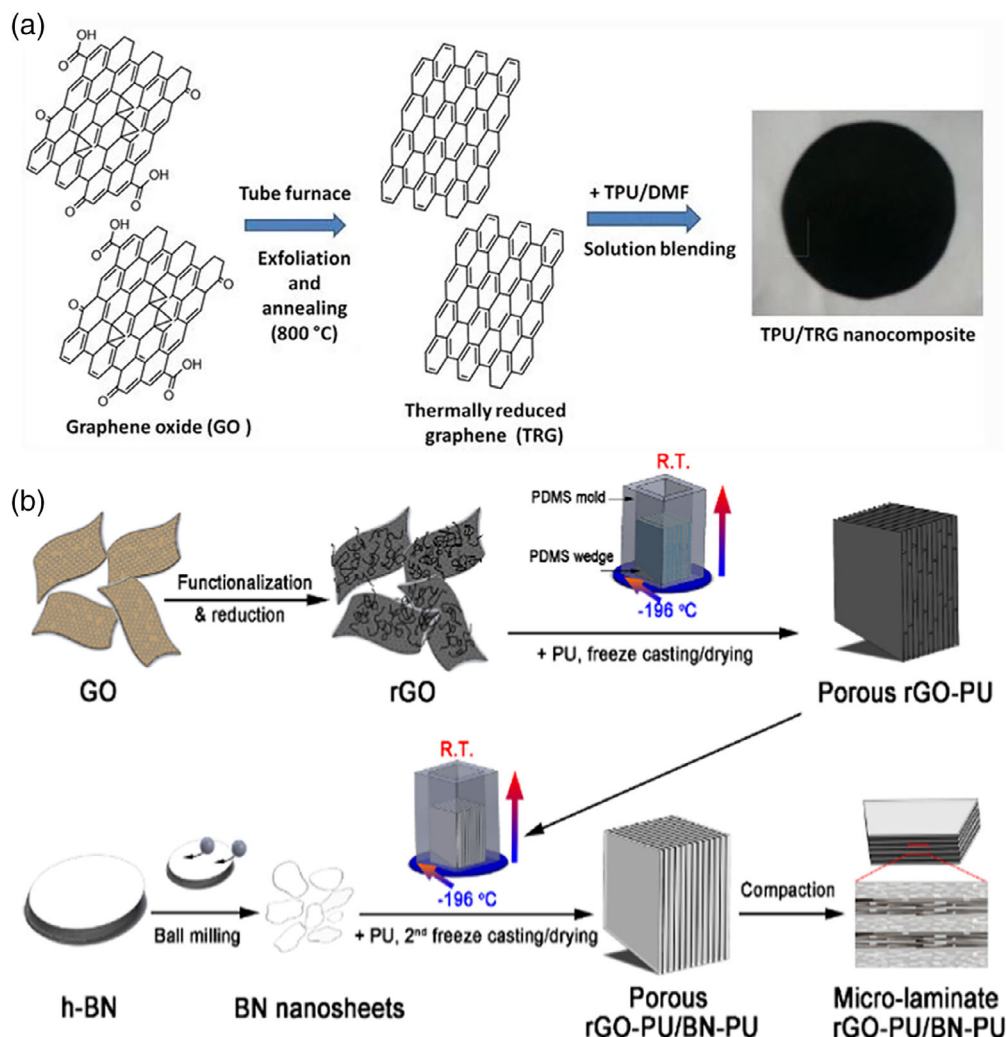
Compared with traditional conductive metallic materials, carbon nanomaterials (such as graphene,^{68,69} graphene oxide,^{70,71} carbon nanotubes⁷²) have promising applications in field of dielectric energy storage devices due to their unique properties, such as light weight, flexible structure, corrosion resistance, and easy processing.

Graphene is a well-known allotrope of carbon consisting of thin layers of sp^2 hybridized carbon atoms arranged in two-dimensional hexagonal lattice. Graphene sheets are widely used in the synthesis of polymer composites for their excellent mechanical, thermal, and electrical properties, which can significantly improve various properties of polymeric materials.^{73–75} The large difference in electrical conductivity between the carbon filler and the insulating polymer matrix leads to interfacial polarization. Well-dispersed graphene-based thermoplastic polyurethane (TPU) nanocomposites were obtained by incorporating thermally reduced graphene (TRG) nanosheets into polyurethane via solution blending,⁷⁶ as shown in Figure 5a. The permittivity of the composites can be significantly improved, owing to the high electrical conductivity of TRGs and polarization of the TRG/TPU nanocomposites. Nevertheless, high energy losses are caused by interfacial polarization on the interfaces between the TRGs and the TPU matrix and the formation of a conductive network approaching the percolation threshold inside the composite.^{77,78} Wu et al.⁷⁹ prepared a graphene/boron nitride-polyurethane composite. This composite was made of reduced graphene oxide (rGO) and boron nitride nanoflakes embedded in a polyurethane (PU) matrix. In

FIGURE 5 Synthesis of carbon composite polyurethane nanocomposites.

(a) Preparation of TPU/TRG nanocomposites.⁷⁶ Copyright 2016, Springer Nature.

(b) Schematic diagram of the synthesis of highly oriented RGO-PU/BN-PU microlaminated composites.⁷⁹ Copyright 2018, American Chemical Society [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]



the composite, the rGO-PU and BN-PU layers were arranged alternately to form highly oriented micro-layered laminates, as shown in Figure 5b, which significantly improved the dielectric properties. The permittivity of this micro-laminated composite is as high as 1084, while the dielectric loss is greatly reduced to below 0.1. This strategy indicates that direct interconnection between small blocks of conductive fillers can be abated by in situ polymerization, growth of inorganic particles, or introduction of insulating polymers on the conductive filler surface. This work points out a direction in developing energy storage capacitors with ultra-high permittivity, low dielectric loss, and improved DBS.

In recent years, CNT-polymer composites have attracted great interest due to their excellent dielectric properties and wide applications in energy storage capacitors and electromechanical actuators. The dielectric properties of CNT composites are limited by the percolation threshold, beyond which the percolation network is formed between conductive carbon nanotubes.^{80,81} A large increase in the permittivity of the composite can

only occur when the volume fraction of the nanofiller is approaching the percolation threshold in the polymer matrix.^{82,83} In the composite, the abrupt reduction of the bulk resistivity near the percolation threshold causes great difficulties in maintaining the permittivity and is accompanied by large conduction losses.^{82,84} It was found that due to their large aspect ratios, carbon nanotubes have a greater advantage in forming percolation networks in polymer matrices at low loads.⁸⁵ Oriented multi-walled carbon nanotubes (MWCNTs)/polyvinyl alcohol (PVA) dielectric composites were made via a simple and controllable film formation technique by in-situ electrospinning.⁸⁶ Composites with high permittivity, low dielectric loss, and high DBS are obtained at low MWCNTs content. The significant improvement in dielectric properties is attributed to the uniform dispersion and orientation of MWCNTs in the matrix. Modified CNT-based composites were also reported. For example, Lu et al.⁸⁷ published a work about a novel composites with multilayered structure: carbon nanotubes (CNT)/cyanate (CE)/hexagonal boron nitride (hBN). This

method achieved an ultra-low dielectric loss and an increased DBS and energy density with only 0.4 wt% of CNTs. Consequently, modified CNT with multilayer structure provides a means of exploring multiphase composites to achieve high permittivity and low dielectric loss.

3.3 | Dielectric polymers based on orientational polarization

Orientational polarization (OP) exists in materials containing unoriented dipoles, responding to a frequency range of about 10^3 – 10^8 Hz. Under an external electric field, dipoles in a polymer will be reoriented according to the direction of the electric field. Water is a typical molecule with strong polarity (with a much larger dipole moment comparing to many other substances) and can be used to understand OP. Due to orientational polarization, the permittivity of pure water at room temperature is as high as 80, and the dipolar relaxation occurs at about 10 GHz, which leads to very low dielectric losses in the power frequency range.⁸⁸ Inspired by the dielectric properties of pure water, people are interested in taking advantage of orientational polarization in polymers to obtain excellent dielectric properties.²⁴ In polymers, dipoles are highly diverse due to differences in polarity of functional groups and chain structures. Various types of polymers with orientational polarization will be discussed below.

3.3.1 | Ferroelectric polymer dielectrics

Based on their responses after removal of the external field, ferroelectric polymer dielectrics can be divided into two types: ferroelectrics and relaxor ferroelectrics. They have aroused broad interest because of their mechanical flexibility and excellent dielectric properties for applications in new energy storage devices, especially thin film capacitors.⁸⁹ Ferroelectric polymers reported recently include polyvinylidene fluoride (PVDF), difluoroethylene-trifluoroethylene copolymer (P(VDF-TrFE)), odd-numbered nylon, and polyurea. Among them, PVDF and copolymer P(VDF-TrFE) are the most widely used.⁹⁰ PVDF-based ferroelectric polymers have potential application prospects in the fields of piezoelectric and thermoelectric energy storage, due to their excellent dielectric and mechanical properties, good chemical resistance, environmental stability, as well as excellent processing characteristics.

Ferroelectric domains exist in ferroelectric polymers, each domain contains dipoles arranged in the same direction, causing spontaneous polarization, in different and domain-specific directions, in the absence of the external

electric field. Under an external electric field, the polarization directions of all the ferroelectric domains tend to become consistent to the direction of the electric field, resulting in OP. Thus, ferroelectric polymers have high permittivity and breakdown strength, which favors their use as dielectric films with high energy storage density. Ferroelectric polymer-based nanocomposites, which combine the good dielectric properties of insulating ceramics and the great mechanical properties of polymers (high strength, flexibility, and ease of processing), have been a focus in the research of dielectric energy storage materials recently.⁹¹ However, due to the presence of large numbers of ferroelectric domains in the composites, their high permittivity and excellent energy density are accompanied by high residual polarization, which leads to hysteresis loss in the process of OP, resulting in low energy storage efficiency under high frequency electric field.⁹² When spontaneous polarization appears in ferroelectric crystals under the external electric field, a depolarization field against the spontaneous polarization will be generated, during which the reorientation has to overcome the “frictional resistance” and lead to dielectric loss. Furthermore, the impurities and other defects would contribute to the friction resistance hence dielectric loss.

In order to solve this problem, the most common strategy is to modify the structure of the polymer, in order to transform a normal ferroelectric polymer into one with relaxor ferroelectric properties. High-energy electromagnetic radiation can impact polymer structures by leading to cross-linking and bond rearrangement. For example, γ -radiation on P(VDF-TrFE) random copolymers was used to obtain relaxor ferroelectric properties and improve dielectric properties.⁹³ In order to transform ferroelectric P(VDF-TrFE) into relaxor ferroelectric phase, its ferroelectric domain stability must be altered. This can be achieved by introducing defects into the polymer structure through electromagnetic radiation, resulting in the increase of amorphous content in the structure and the formation of nanoscale ferroelectric domain. The research shows that with the increase of irradiation intensity, the ferroelectric domain in P(VDF-TrFE) can be reduced to 5 nm.⁹⁴ The relaxor ferroelectric behavior of P(VDF-TrFE) under radiation can be observed, which shows that the Curie temperature disappears and the residual polarization decreases. Besides physical methods, relaxor ferroelectric properties can be obtained by introducing a third monomer into the polymer main chain to allow substantial changes in the polymeric structure by controlling the chain spacing of the copolymer. For example, chlorofluoroethylene (CFE) and chlorotrifluoroethylene (CTFE) were added to P(VDF-TrFE) to synthesize terpolymers.^{95,96} The unique ferroelectric nanodomain structure contained in the relaxor

ferroelectrics gives them desirable dielectric properties as well as low hysteresis losses in the potential-shifted field strength loops because of the addition of the third monomer created longer main chains and increased the polarity of functional groups on the main chains.

To further reduce the dielectric loss factor of the polymer, attention has been paid to the paraelectric polymer, which has molecular chains and dipoles that can easily rotate according to the direction of the applied electric field and thus do not form ferroelectric nanodomains during polarization under high frequency alternate electric field.⁹⁷ Noticeably, the high conductivity of electrons in the polymer matrix confined the improvement of the dielectric properties. Generally, blending polymer nanodielectrics are able to modify the dielectric polarization of ferroelectric polymers more easily than modifying the structure of polymer molecules.^{32,98} Although modifying the molecular structure of ferroelectric polymers can effectively reduce the energy loss of materials, this faces many difficulties. Thus, it is necessary to find a more appropriate strategy to further pursue the ideal ferroelectric materials with high permittivity and low loss.

3.3.2 | Dipolar glass polymers

Dipolar glass polymer (DGP)²⁶ is a kind of polymer with a high glass transition temperature, whose high permittivity is achieved by utilizing the OP, the dominant type of polarization under glass transition temperature (T_g). Compared with ferroelectric polymers, DGP has a low dielectric loss, because electrical conduction of charges is suppressed at glassy state.⁹⁹ High T_g makes it a desirable material for high temperature, high energy density, and low loss electric energy storage.

For DGP, individual dipolar groups are confined but mobile within the free volume of the polymer matrix under T_g . Note that DGP is different from polar molecule-doped glassy polymers (PMDGP),^{100,101} which contains doping polar molecules too large to rotate freely in the polymer matrix below T_g . Moreover, when the doping concentration is high, these dipolar molecules tend to aggregate and separate from the polymer phase. DGP does not suffer from this kind of phase separation. DGP's dipolar groups are much smaller than the doping polar molecules in PMDGP. Due to the suppressed mobility of the glassy polymer chains below T_g , DGPs exhibit low conductive loss and low polarization loss. Dielectric loss is reduced significantly because of weak interaction between dipolar groups, in the absence of translational motions in the glassy state, and lack of spontaneous ferroelectric domains.

According to the positions of the dipolar groups in DGPs, they are generally divided into main-chain DGPs and side-chain DGPs. Crown ether groups were introduced in the polyimide (PI) main-chain to increase the polarization of the molecular chain.¹⁰² The PI films prepared have high permittivity and low dielectric loss, while still maintaining its mechanical and thermal properties. The origin of ferroelectricity in glassy nylon using nearly 100% amorphous aromatic nylon was investigated.¹⁰³ In this case, it is difficult to distinguish OP from atomic polarization under high-frequency electric fields, DGPs are mainly used in the medium- and low-frequency range, in which OP dominates.

Since the rotation of dipoles on the main chain is rather limited, adding dipoles to the side chains can improve OP of DGPs.⁹⁹ Because the resistance of the side-chain rotation is greatly reduced, the side-chain dipoles are more desirable than the main-chain dipoles. The dipoles on side-chain of DGP, such as those formed by carbonyl, cyanomethyl (cyanoethyl), and methyl sulfonyl groups, are capable of rotating nearly 180°, with small hindrance, promoting OP and reducing dielectric loss. A new class of high- T_g dipolar polymers were synthesized by post-polymerization functionalization based on sulfonated poly(2,6-dimethyl-1,4-phenoxy) (SO₂-PPO),¹⁰⁴ as shown in Figure 6a. The OP of SO₂-PPO is enhanced by the effective rotation of the highly polar methyl sulfonyl side-chain group below T_g , resulting in a high permittivity and a very low loss factor ($\tan\delta = 0.003$). Furthermore, monosulfonyl (CH₃SO₂—) and disulfonyl [CH₃SO₂(CH₂)₃SO₂—] groups were grafted on polyepichlorohydrin(mono-substituted) and poly(3-bis [chloromethyl] oxane) (double-substituted), respectively, to change the dipole density and dipole arrangement of the side chains,¹⁰⁵ as shown in Figure 6b. Due to the OP of high polar sulfonyl groups, these DGPs exhibit higher permittivity and lower loss factor ($\tan\delta$ of 0.003–0.02) in the glassy state. This is because the resistance of the side-chain dipole is greatly reduced. It seems that DGP are advantageous dielectric polymer materials to achieve high permittivity and low dielectric loss.

3.4 | Dielectric polymers induced by relaxation ionic polarization

RIP in dielectrics is the ion diffusion movement caused by the difference in ion concentration at the metal electrolyte interface of the electrode, so as to form an electric double layer (EDL). Among polymeric dielectrics whose dominant polarization is RIP, the most common case is ion-gel. In this review, we focus on polymeric ion-gels formed by ionic liquid immobilized in a polymeric matrix

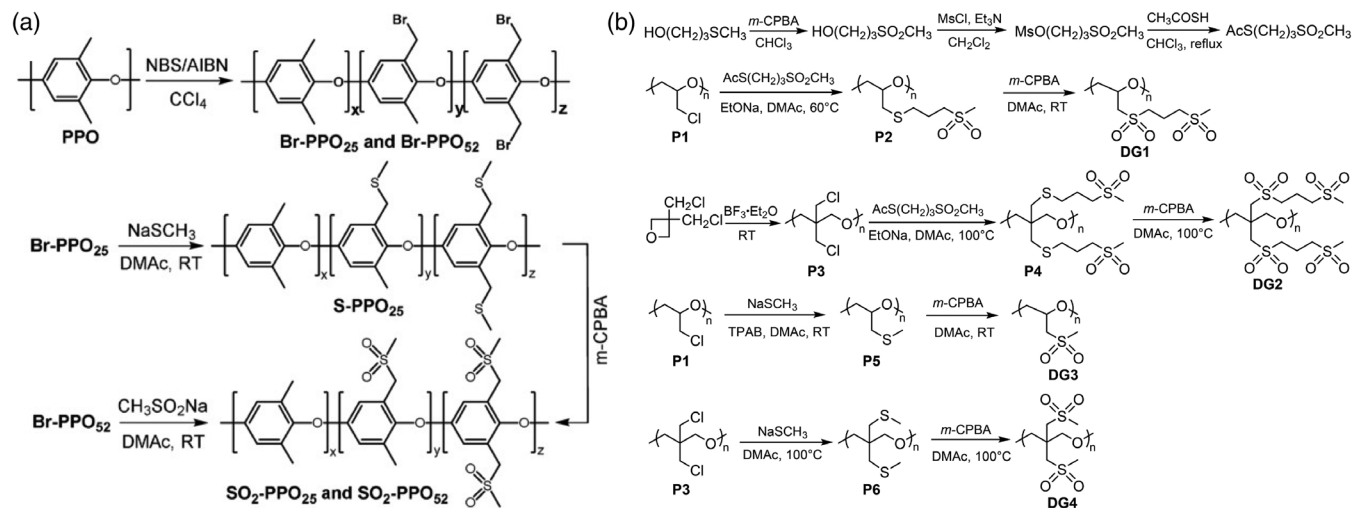


FIGURE 6 Schematic diagram of the synthesis of dipolar polymers. (a) Two synthetic routes of sulfonated SO₂-PPO.¹⁰⁴ Copyright 2018, John Wiley and Sons. (b) Synthesis of four dipolar glass polymers (DGPs), DG1-DG4.¹⁰⁵ Copyright 2018, American Chemical Society

in which the polymer chains are interconnected or entangled to form a 3D network structure. In energy storage devices, ion-gels were used as dielectrics, functioning as a gate insulator for electrolyte gated transistors (EGTs).¹⁰⁶ The structure comprises an ionic adsorption layer close to the polymer dielectric interface, a diffusion layer between the interface and the interior of the matrix, and a conductive electrode. The assembled supercapacitor or battery stores energy through an ionic polarization process involving EDLs.^{107,108}

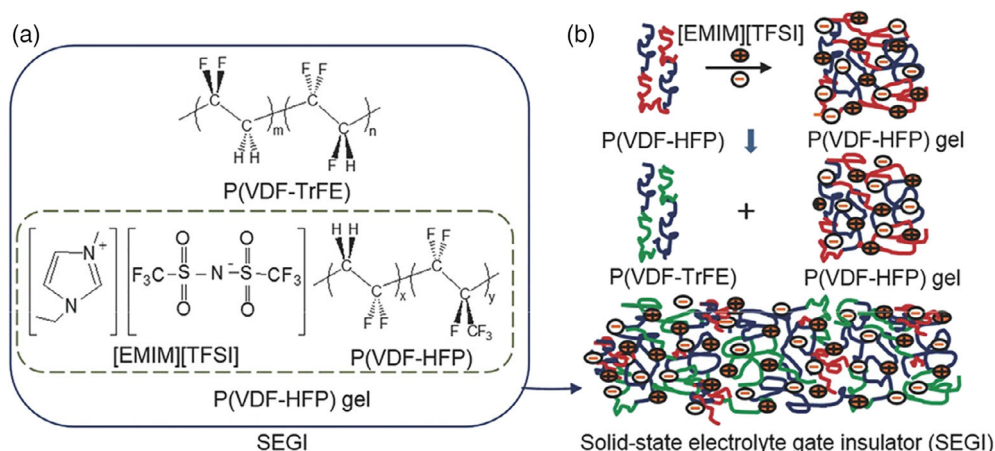
The so-called RIP in polymers (different from IIP in ionic crystals) is caused by the long-distance (>10 nm) migration of ions into the EDLs formed between polymer-electrode interface and the interior of the matrix when polymer dielectrics are used in energy storage devices. The process of IRP involves the generalized physical migration of ions on the outer surface of polymer dielectrics and the diffusion layer, which usually occurs between a fraction of a second and a few hours, and the frequency is about 100 Hz. Under the electric field, ions with excessive transition in the direction of electric field form electric potential difference due to uneven distribution of dielectric charge, which bring about polarization.

Recently, the successful use of ion-gels as polymeric dielectrics has intrigued great interest in its research.^{109,110} Most ion-gels consist of colloidal block copolymers, which form a cross-linked network structure, providing channels for movement of ions and high tensile strength for the materials. Ionic liquids mixed in copolymers are strong electrolytes in molten state. Many active sites on the functional groups of the copolymers are used to form coordination bond of cations and anions dispersed in the ion-gels on the polymeric

chains. Under an external electric field, the migration of anions and cations to opposite directions results in an uneven charge distribution inside the two EDLs (one on each end of the ion-gel). The EDLs play a pivotal role in the electrical property of ion-gels, especially in elevating permittivity via IRP between the two interfaces (formed by the two EDLs) and in enhancing its performance in energy storage.³² An ion-gel dielectric with a specific capacitance up to two to three orders of magnitude higher than that of conventional dielectrics was reported.¹¹¹ The high capacitance of ion-gels arises from the formation of nanometer-thick EDL structure with tightly arranged free ions at the interface.^{112,113} However, in the process of ion conduction, free ions tend to aggregate into clusters in the ionic liquid and electrolyte(s), thereby increasing the formation time of EDL in the ion gel.

To improve the dispersion of anions and cations during polarization and promote the formation of EDLs, the ions can be mixed with an additional polymeric medium to prevent aggregation of ions (due to attraction between cations and anions) in the process of ion conduction triggered by polarization after applying an external electric field.¹¹⁴ Nketia-Yawson et al.¹¹⁵ reported a solid electrolyte gate insulator (SEGI), which is composed of a high permittivity polymer poly(vinylidene fluoride-trifluoroethylene) [P(VDF-TrFE)] and a controlled mixture consisting of an ionic gel based on poly(vinylidene fluoride-co-hexafluoropropene) [P(VDF-HFP)] and 1-ethyl-3-methylimidazolium bis(trifluoromethylsulfonyl) imide ([EMIM][TFSI]), as shown in Figure 7. The outcome reveals that the dispersion of ions and the density of charge carriers in the dielectric layer of the substance are significantly improved, which is conducive to promoting the formation of EDL.

FIGURE 7 Schematic diagram of the chemical structure of the substances used and the formation of solid state electrolyte grid insulator (SEGI). (a) the molecular structures of the blends include P(VDF-TrFE), [EMIM][TFSI], and P(VDF-HFP). (b) Diagram of the formation mechanism of SEGI.¹¹⁵ Copyright 2017, John Wiley and Sons [Color figure can be viewed at wileyonlinelibrary.com]



While ion gels can make a great contribution to the dielectric capacity of polymer composites, the losses in ion gels become considerable at frequencies of the applied electric field above 10^4 Hz.³² The root of the above problem is that the ions in the EDL migrate slowly and are unable to keep up with the electric field at high enough frequencies. Obviously, significant ion transportation on the interface of EDL between the polymer dielectric and the electrode should be avoided. Therefore, in the preparation of polymer dielectrics, multilayer films are used to reduce the high dielectric loss caused by ion migration. One publication covered the high temperature polycarbonate (HTPC)/polyvinylidene fluoride (PVDF) multilayer membranes prepared by multilayer extrusion technology, and its restricted ion transportation under the action of high AC electric field was studied by direct analytical simulations.¹¹⁶ Such polarized multilayer membrane effectively reduces the ionic conduction loss of the material, decreases the leakage current, and improves the DBS. It is shown that the increased number of layers in multilayer structure, especially the formation of multilayer nanofilms, can more effectively inhibit the long-distance transport of ions. Thus, for dielectric polymers, multilayers are a promising way to reduce the conduction loss of impurity ions in dielectrics.

Although multilayer films can be used to effectively reduce the energy loss of materials, ion-gel polymer dielectric materials are still not suitable for high frequency film capacitors. Further exploration is needed to deal with the dielectric loss caused by ion conduction. In an inspiring study, the ion conduction loss of polysulfone (PSF)/PVDF multilayer films was suppressed by a simple melt recrystallization method, and the mechanism of ion conduction inhibition was revealed by various techniques.¹¹⁷ In this research, a multi-layer extrusion technology with controlled cooling rate is proposed, which has a significant effect on reducing the high temperature dielectric loss of dielectric films. In a word, polymer

dielectrics based on RIP are not ideal for low loss energy storage equipment requiring rapid charge and discharge.

4 | POLYMER DIELECTRIC MATERIALS WITH SYNERGISTIC EFFECT OF MULTIPLE POLARIZATIONS

As we mentioned earlier, polar polymers based on OP are promising for improving permittivity and reducing energy losses. However, the limitation is that polar polymers are potentially hygroscopic and hence susceptible to ionic impurities, which causes the polymer dielectric loss due to conductivity.

In recent years, a large number of studies have focused on polymer dielectric systems with multiple polarizations, the most common of which are OP and IFP-based systems. A novel low-loss double (double chain) block copolymer, including ion-conjugated conductive chain segments with narrow band gaps and insulating chain segments with wide band gaps, using sequential complex decomposition method was synthesized.¹¹⁸ The copolymer was found to have a special nanostructure, with multiple polarizations such as EP, OP, and IFP, all of which greatly improved the dielectric properties of the polymer films. Rani et al.¹¹⁹ prepared nanocomposite films with different contents of carbon black nanoparticles (CBNPs) reinforced with polyvinyl alcohol (PVA)/polyethylene glycol (PEG) by a solution-casting method, as shown in Figure 8. High concentration (25 wt%) of CBNPs exhibited a high permittivity of 375.1. Nevertheless, the dielectric loss of the nanocomposite films was correspondingly high due to multiple factors such as OP, IFP, and electrical conductivity.

Research on reducing conductive loss by multilayer membrane structures was mentioned earlier. To cope with the issues of dielectric loss of the systems mentioned above, polar polymers and nonpolar polymers with OP

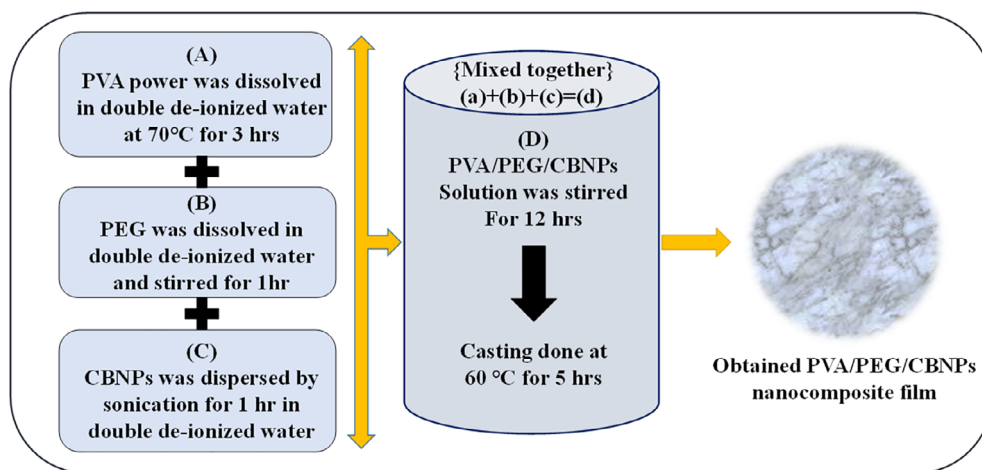


FIGURE 8 Synthesis process of PVA/PEG/CBNPs co-blended nanocomposite films¹¹⁹ [Color figure can be viewed at wileyonlinelibrary.com]

dominance can be used to build systems with multilayer-membrane structure and multi-polarization coaction. Multilayer membranes contain alternating polymers include those with remarkable permittivity (e.g., PVDF and ferroelectric terpolymers) and those with high breakdown strength/low dielectric loss (e.g., polysulfone and polycarbonate). Numerous studies have found that multilayer films are unique one-dimensional systems with tailored materials, layer thicknesses, and interfaces.¹²⁰ By minimizing the unfavorable polarization and enhancing the favorable polarization, the dielectric loss caused by impurity and ion migration during polarization can be reduced. In the field of high permittivity and low dielectric loss polymer materials, multilayer films are expected to be the ideal dielectric materials for future polymeric thin film capacitors.

5 | APPLICATIONS AND FEASIBILITIES

The above sections analyzed the preparation and application scenarios of dielectric polymers based on different polarizations. We further discuss relevant mechanisms, with examples of regulating dielectric properties of the polymers, and feasibilities of their applications.

Dielectric properties of polymers depend on the frequency of external electric field. While permittivity is closely related to types of polarization depending on frequency, dielectric loss is essentially caused by loss of energy due to charge movement or dipole reorientation, which also depends on the frequency of external electric field. When low frequency ($< \sim 10^2$ Hz) electric field is applied, almost all types of polarizations can keep up with the change (alternation) of the electric field, which permits high degree of polarization and hence high permittivity. In the meantime, reversing of charge flow and/or flipping of

dipoles are less frequent under low frequency field, leading to low dielectric loss. Under high frequency ($> \sim 10^8$ Hz) electric field, only EP catches up with the change of electric field and therefore contributes to permittivity and dielectric loss. Without contribution from other types of polarization, the overall permittivity and dielectric loss of the material are both low generally. Since high permittivity and low dielectric loss can be realized under low frequency but unrealistic under high frequency, researchers should pay attention to optimization of the two in the intermediate frequency range ($\sim 10^2$ – $\sim 10^8$ Hz).

For polymer dielectrics, dielectric properties are affected by not only external factors (such as frequency of electric field and temperature), but also regulatable internal factors (such as the filler loading level and interfacial morphology of the fillers). For example, Xie et al.¹²¹ synthesized two types of polymer composites and studied their dielectric properties under a range of frequencies. Type I composites (PVDF/BT) are made of PVDF matrix filled with different loadings of BaTiO₃ (BT) particles. Type II composites are made of BaTiO₃ particles encapsulated by polydopamine (PDA), as fillers (PDA@BT), with different loadings, for the PVDF matrix. Figure 9 includes diagrams showing dependence of permittivity and loss factor ($\tan\delta$) on frequency in the intermediate range (10^2 – 10^6 Hz). Note that Type II composites are marked as c-PVDF/PDA@BT in Figure 9c,d, where c-PVDF means cross-linked PVDF. In this case, PDA not only encapsulated BT particles, but also facilitated cross-linking within the Type II composites. In this system, the increase of filler loading at the same frequency raises both the permittivity and the dielectric loss. Figure 9b,d show that different filler types and loadings impact $\tan\delta$ differently in terms of extent, the impact is synergistic and frequency-dependent. For example at the same loading and at 10^2 Hz, the modified filler reduces $\tan\delta$ to a lower value (0.2), comparing to unmodified filler BT.

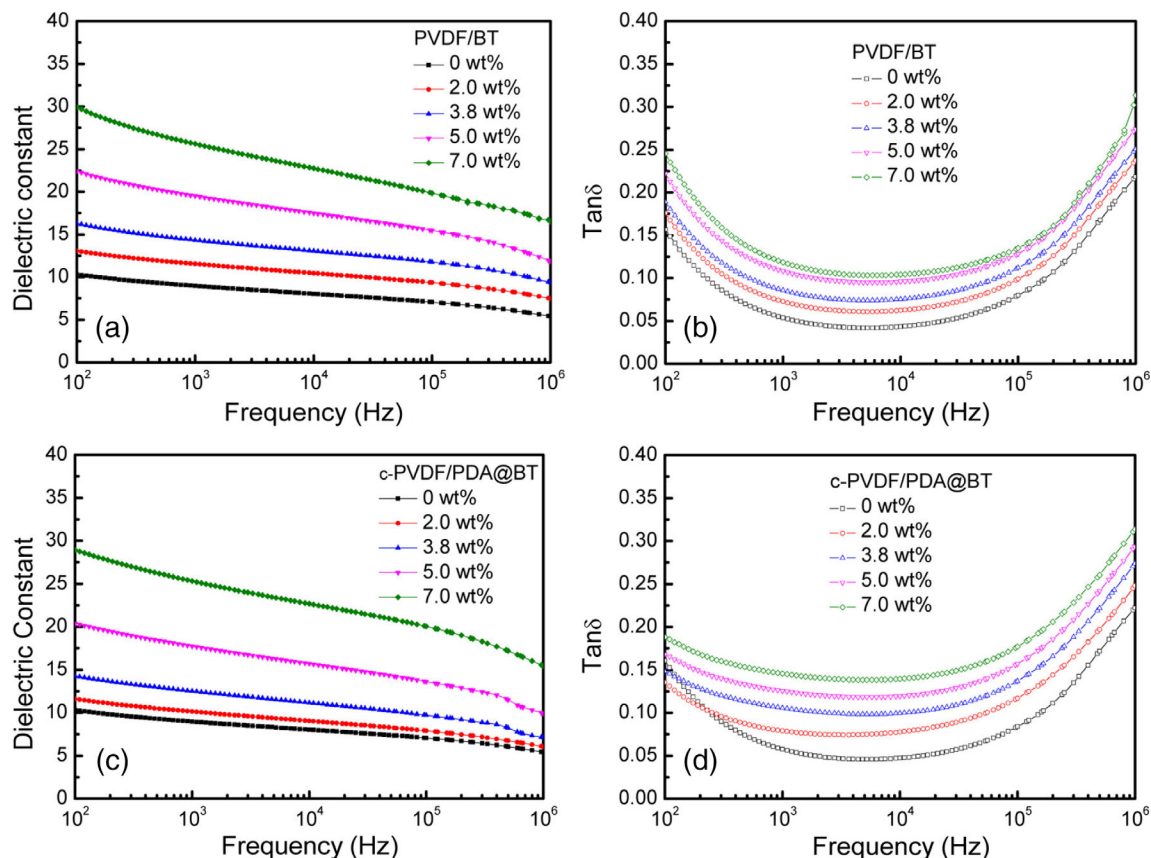


FIGURE 9 Frequency dependence of permittivity (a and c) and dielectric loss (b and d) of polymer nanocomposite with different loadings of BT and PDA@BT particles under room temperature. Copyright 2017, American Chemical Society [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/doi/10.1002/app.52367)]

TABLE 1 Dielectric properties of different polymer composites

Polymer dielectrics	Filler	Matrix	Optimal loading	Frequency	ϵ	$\tan \delta$	Reference
Ceramic-polymer composites	Ba _{0.5} Sr _{0.5} TiO ₃	P(VDF-CTFE)	40 vol.%	100 Hz	38.4	0.02	49
	BaTiO ₃	PVDF	30 vol.%	1 kHz	27	0.04	53
	BaTiO ₃	PMMA/Epoxy	30 vol.%	100 Hz	375	0.02	55
Metal-polymer composites	Zn-ZnO	PVDF	14 vol.%	1 kHz	60	0.07	64
	EGaIn	PBMA	40 vol.%	1 kHz	23	0.07	67
Carbon nanofiller-polymer	rGO /BN	PU	0.73 vol.%/3.85 vol.%	1 kHz	1084	0.091	79
	MWCNTs	PVA	4.0 vol.%	100 Hz	105	0.8	86
	CNTs/BN	CE	0.4 wt.%/20 wt.%	100 Hz	323	0.049	87

The dependence of permittivity and dielectric loss (indicated by $\tan \delta$) on frequency can be mechanistically explained in general. For permittivity, when the frequency is less than 10^2 Hz, both IFP and OP contributes to the dielectric properties, leading to a relatively high permittivity. With the increase of frequency ($>10^2$ Hz), the contribution of IFP diminishes and only OP contributes to polarization, hence the overall polarization decreases,

therefore, the permittivity shows a downward trend (Figure 9a,c). For the loss factor, when the frequency is less than 10^2 Hz, the dielectric loss is mainly contributed by IFP. With increase of frequency, the loss factor first decreases to a minimum value at about 10^4 Hz and then keeps increasing till 10^6 Hz (Figure 9b,d). The decreasing trend from 10^2 to 10^4 Hz can be explained by diminishing contribution of charge movements by IFP and OP as

frequency increases. Beyond $\sim 10^4$ Hz, $\tan\delta$ starts to increase because of the lagging of dielectric response of dipoles at higher frequency.

We examined a few more cases in the same research area.^{49,53,55,64,67,79,86,87} These examples show similar trends as we described above. We listed key parameters

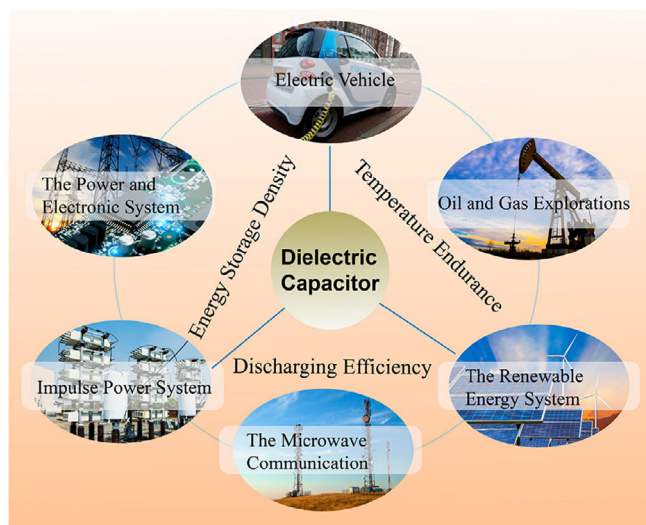


FIGURE 10 Application fields of dielectric capacitors.¹²⁵ Copyright 2021, American Chemical Society [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

in Table 1, to demonstrate the relations of these factors. Filler, Matrix, optimal loading, and frequency were listed with correspondent values of permittivity and $\tan\delta$. Details can be found in the references listed in the table.

In principle, optimal permittivity and minimal energy loss depend on characteristics of both the filler(s) and the polymer(s), including the type, structure, loading of the filler(s), and the filler-polymer matching. Among the research of polymer dielectrics, efforts have been made to regulate/optimize dielectric properties. For examples, porous dipole polymers were prepared to promote the drag-free rotation of side groups on the molecular chains,¹²² a nano filler with core-shell structure was designed to prevent the electron transmission between the two phases and inhibit the percolation phenomenon,¹²³ and a transition interlayer was introduced to realize the dual action on the filler-interlayer and interlayer-matrix interfaces.¹²⁴

Although polymer dielectrics have all the advantages mentioned previously, their energy density values do not meet the requirements for high-power energy storage systems. Currently, they are typically used in small electronic components such as polymer film capacitors and flexible wearable electronic devices. Dielectric capacitors have been widely utilized in many fields, such as microwave communication, pulse energy system, sustainable energy system, and electric vehicles,¹²⁵ as shown in

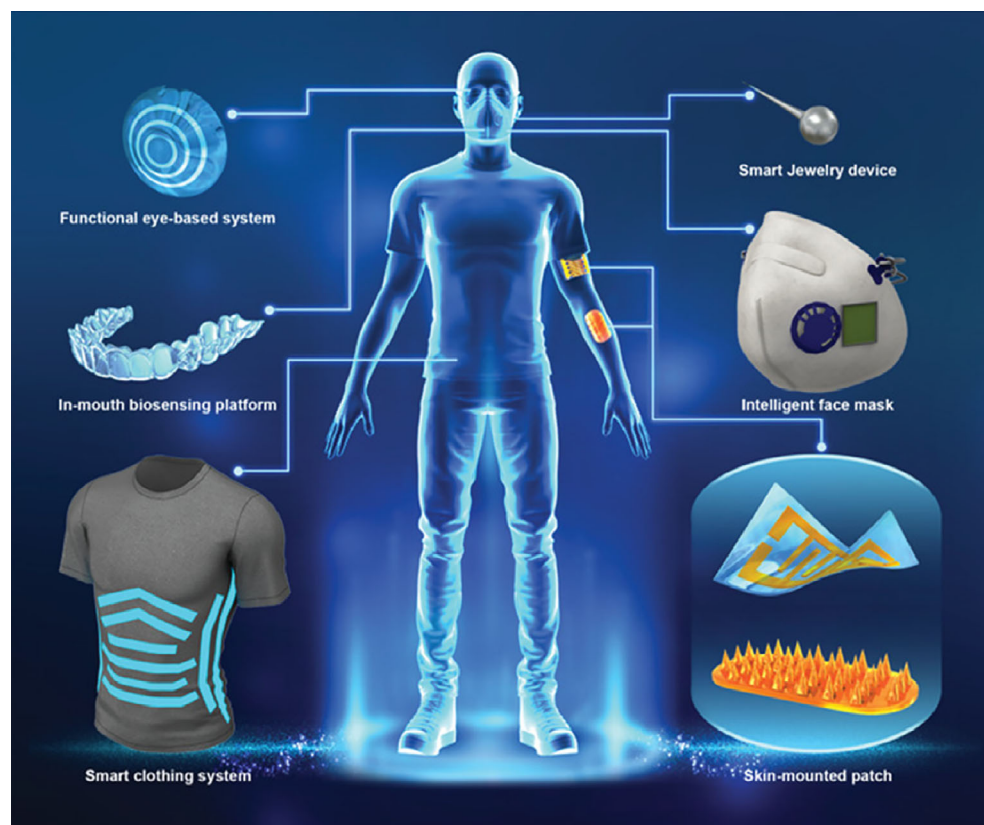


FIGURE 11 Intelligent wearable device and its application.¹²⁷ Copyright 2021, John Wiley and Sons [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com/terms-and-conditions)]

Figure 10. Polymer film capacitors have the advantages of low dielectric loss, stable current density, and long service life.¹²⁶ With the development of effective production technology and advanced electronic equipment, polymer film capacitors with excellent energy storage capacity show promising prospects in the field of energy conversion.

Small, intelligent, and life-changing wearable devices are gaining popularity and widely used in many fields, such as health monitoring and biological signal detection. These devices often require miniaturization, light weight, low loss, sensitive feedback, accurate response, and good wearability. A variety of products and prototypes based on polymer dielectrics emerge, products such as skin patches, smart clothes, functional contact lenses, smart braces, smart jewelry, and smart masks are shown in Figure 11.¹²⁷

6 | CONCLUSIONS AND OUTLOOK

With electronic devices trending toward miniaturization and high performance, there is an urgent need for the development of dielectric materials with high energy storage density. Dielectric polymers have become the first choice for the application of smart electricity electronics and pulsed power, with great potentials in the field of energy storage and flexible sensing.

The combination of polymer and ceramics, carbon materials or conductive metals can not only effectively improve the energy storage efficiency of polymer materials, but also optimize the mechanical properties of materials to a certain extent. However, most of the dielectric polymer composites still have great limitations in the control of energy loss. Consequently, the application of multilayer thin film technology in multipolarization systems can also effectively reduce the dielectric loss of polymer materials, which has aroused widespread interest among researchers. In particular, this new technology has a significant effect on the improvement of the dielectric properties of the polymer itself and of the polymer composites. Overall, dielectric properties can be improved by modifying fillers and/or polymer matrices, to change the internal structure of the composites. In this review, applications of polymer dielectrics in energy storage devices are generally aimed at small energy storage or sensing devices, such as polymer film capacitors and wearable devices in the biomedical field where the demand is growing. To achieve high efficiency, high energy density, and miniaturization, it is necessary to further explore the correlation between composition/structure and dielectric properties.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

AUTHOR CONTRIBUTIONS

Jie Qiu: Writing – original draft (equal). **Qun Gu:** Writing – original draft (equal). **Ye Sha:** Investigation (supporting). **Yang Huang:** Investigation (supporting). **Meng Zhang:** Writing – review and editing (lead). **Zhenyang Luo:** Writing – review and editing (lead).

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